



राष्ट्रीय भू-सूचना विज्ञान
एवं प्रौद्योगिकी संस्थान
भारतीय सर्वेक्षण विभाग
विज्ञान और प्रौद्योगिकी विभाग

National Institute of Geo-Informatics
Science & Technology
Survey of India
Department of Science & Technology

STANDARD OPERATING PROCEDURE
for
Field Operation of
NRTK Method using
(Trimble R980 GNSS & Trimble Access)

Course Name				
Training Module Name	NRTK Method of GNSS Observations			
Description	NRTK Method using Trimble R980 & Trimble Access			
Created By	N SHASHI KIRAN, Surveyor, Faculty of Geodesy, NIGST			
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1	N SHAHSI KIRAN , Surveyor			

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1. Introduction

This Standard Operating Procedure (SOP) provides step-by-step instructions for conducting NRTK surveys using the Trimble R980 GNSS receiver and Trimble Access software. Prepared by NIGST, this SOP aims to guide users through essential functions, from instrument setup to data export, ensuring accurate data collection and efficient workflow. As users gain familiarity with the equipment, they are encouraged to explore advanced features to further enhance survey productivity.

2. Objective

The main objectives of conducting an NRTK GNSS survey using the Trimble R980 receiver are:

- To achieve high-precision, centimeter-level real-time positioning using the Trimble R980 GNSS receiver
- To utilize network-based corrections (VRS/MAC/FKP) for improved accuracy over large areas
- To eliminate local base station setup, reducing field time and operational complexity
- To minimize distance-dependent and atmospheric errors through NRTK services
- To ensure fast initialization and stable RTK fixes with the R980's advanced GNSS engine
- To support multi-constellation and multi-frequency tracking for better reliability in challenging environments
- To improve productivity and data quality in surveying, mapping, construction, and GIS applications
- To enable continuous real-time data collection using mobile internet connectivity.

3. Note on RTK

Network RTK (NRTK) is an advanced GNSS positioning technique that uses a network of continuously operating reference stations (CORS) instead of a single base station. The data from these reference stations are processed at a control center, which models atmospheric errors (ionospheric and tropospheric) and satellite-related errors over the entire network. Correction data are then transmitted to the rover in real time via the internet (GPRS/4G/5G).

➤ Key Features

NRTK provides high-accuracy real-time positioning, typically achieving centimeter-level accuracy in both horizontal and vertical components. Common NRTK methods include VRS (Virtual Reference Station), FKP (Flächen-Korrektur-Parameter), and MAC (Master–Auxiliary Concept).

Key advantages of NRTK include uniform accuracy over large areas, no need to set up a local base station (Like in RTK), improved reliability, and reduced baseline-dependent errors. Due to these benefits, NRTK is widely used in land surveying, mapping, construction, deformation monitoring, and GIS applications.

➤ **Applications**

NRTK is widely used for topographic mapping, engineering surveys, boundary surveys, infrastructure planning, monitoring, and control point establishment. It ensures standardized, repeatable, and highly accurate GNSS data, making it essential for projects such as AMRUT, NAKSHA, ICZM, and state resurvey programs.

➤ **Limitations**

Requires reliable internet/mobile data connectivity for real-time corrections. Performance depends on the density and geometry of the reference station network. Accuracy degrades outside the network coverage area. May be affected by server outages or correction service downtime. Subscription or service cost is usually involved. Still susceptible to multipath and signal obstruction in urban/forested areas. Less effective during severe ionospheric disturbances (e.g., high solar activity)

4. Hardware & Software

4.1 Hardware

➤ **Trimble R980 GNSS Receiver**

- a) Multi-frequency, multi-constellation GNSS receiver (GPS, GLONASS, Galileo, BeiDou, QZSS, NavIC-ready) / Main Rover unit
- b) Rover should be compatible IMU-based tilt compensation for tilt observations

➤ **GNSS Antenna (Integrated)**

- a) High-precision antenna built into R980 for accurate phase tracking

➤ **Survey Controller / Data Collector**

- a) Trimble TSC7 / TSC5 / TCU5
- b) Runs field software and controls the receiver

➤ **Communication Devices**

- a) Internet modem / SIM card (4G/5G) for receiving NRTK corrections over mobile network

➤ **Bipod with Survey Pole**

- a) For rover observations

➤ **Batteries & Chargers**

- a) Rechargeable batteries for the R980 and controller
- b) Portable power bank or vehicle charger for extended field work.

➤ **Cables & Accessories**

- a) Cables, mounts, protective cases, and data transfer media.

4.2 Software

➤ **Trimble Access (Field Software)**

Field survey software for data collection, instrument control, and RTK

➤ **Trimble NTRIP Client**

Software or firmware in R980 to receive real-time NRTK corrections from NTRIP casters.

➤ **NTRIP Caster / Subscription Service**

Access to an NRTK network correction service (e.g., CORS service provider);
Credentials (username, password, server URL)

5. Pre requisites

➤ **GNSS Equipment**

- NRTK-capable GNSS receivers (Rover) .
- Multi-frequency, multi-constellation support for reliable fixes.

➤ **NRTK / NTRIP Subscription**

- Active access to a CORS/NRTK network (VRS / MAC / FKP)
- Server address, port, mount point, username, and password.

➤ **Reliable Internet Connectivity**

- SIM card with stable 4G/5G mobile data coverage in the survey area.

➤ **Field Controller**

- Compatible controller (e.g., Trimble TSC7 / T10) with **Trimble Access** installed.

➤ **Correct Coordinate System**

- Proper **datum, projection, geoid model**, and transformation parameters loaded in Trimble Access.
- **GNSS Satellite Visibility**
- Clear sky view to track **multi-constellation signals** (GPS, GLONASS, Galileo, BeiDou, NavIC)
- **Power Supply**
- Fully charged batteries for receiver and controller; spare batteries recommended for extended observations in field.
- **Operator Knowledge**
- Understanding of NRTK principles and its methods.
- Correct handling of GNSS equipment and settings

6. Roles & Responsibilities

During an NRTK GNSS survey, clear roles and responsibilities are essential to ensure accuracy, efficiency, and data quality.

A field Observer shall:

- Plan the survey methodology and accuracy requirements.
- Ensure availability of NRTK subscription, hardware, and software.
- Manage CORS/NRTK service access (NTRIP credentials, mount points).
- Ensure correct coordinate systems, datum, and geoid models.
- Monitor network performance and resolve connectivity issues.
- Set up and operate the Trimble R980 rover.
- Check RTK fix status, PDOP, satellites, and QC parameters.
- Maintain survey equipment and field records.
- Ensure stable mobile data connectivity and device configuration.

7. Specifications of Trimble R980 GNSS Receiver

The Trimble R980 GNSS system is a premier solution for land surveyors, offering exceptional accuracy and efficiency. Featuring the Trimble ProPoint® GNSS engine and Trimble Inertial Platform™ (TIP™) IMU-based tilt compensation, users can collect precise data in challenging environments without leveling the pole. Below are the specifications of Trimble R980 GNSS Receiver:

- **Performance specifications**
 - **GNSS TECHNOLOGY**
 - 1) Constellation agnostic, flexible signal tracking, improved positioning in challenging environments and inertial measurement integration with Trimble ProPoint® GNSS technology.

- 2) Increased measurement and stakeout productivity and traceability with Trimble Inertial Platform™ (TIP™) technology IMU-based tilt compensation.
- 3) Dual Trimble Maxwell™ 7 Custom GNSS chips with 672 channels.
- 4) Trimble EVEREST™ Plus multipath signal rejection.
- 5) Trimble IonoGuard™ technology for mitigation of ionospheric GNSS signal disruptions.
- 6) Trimble CenterPoint® RTX correction service is activated and ready to use for the initial 12 months.
- 7) Spectrum Analyzer to troubleshoot GNSS jamming.
- 8) Digital Signal Processor (DSP) techniques to detect and recover from spoofed GNSS signals.

- **SATELLITE TRACKING**

GPS: L1C, L1C/A, L2C, L2E, L5

GLONASS: L1C/A, L1P, L2C/A, L2P, L3

SBAS (WAAS, EGNOS, GAGAN, MSAS, SDCM): L1C/A, L5

Galileo: E1, E5A, E5B, E5 AltBOC, E62

BeiDou: B1I, B1C, B2I, B2A, B2B, B3I

QZSS: L1C/A, L1S, L1C, L2C, L5, L6

NavIC (IRNSS): L5

L-band: Trimble RTX® Corrections

➤ **Positioning performance**

- **STATIC GNSS SURVEYING**

	Horizontal	Vertical
High-Precision Static	3 mm + 0.1 ppm RMS	3.5 mm + 0.4 ppm RMS
Static and Fast Static	3 mm + 0.5 ppm RMS	5 mm + 0.5 ppm RMS

- **REAL TIME KINEMATIC SURVEYING**

	Horizontal	Vertical
Single Baseline < 30 km	8 mm + 1 ppm RMS	15 mm + 1 ppm RMS
Network RTK	8 mm + 0.5 ppm RMS	15 mm + 0.5 ppm RMS
RTK start-up time for specified precisions	2 to 8 seconds	

- **TRIMBLE INERTIAL PLATFORM (TIP) TECHNOLOGY**

	Horizontal
TIP Compensated Surveying	RTK + 3mm + 0.15mm/°tilt (up to 40°) RMS
	RTX + 3mm + 0.15mm/°tilt (up to 40°) RMS

- **TRIMBLE RTX CORRECTION SERVICES**

CenterPoint RTX	Horizontal	2 cm RMS
	Vertical	3 cm RMS
	Convergence time for specified precisions in Trimble RTX Fast regions	< 1 min
	Convergence time for specified precisions in non Trimble RTX Fast regions	< 3 min
	Quick Start convergence time for specified precisions	< 1 min

- **TRIMBLE xFill**

	Horizontal	Vertical
TRIMBLE xFill	RTK + 10 mm/minute RMS	RTK + 20 mm/minute RMS

➤ **Hardware**

- **PHYSICAL**

Dimensions (W×H)	11.9 cm x 13.6 cm (4.6 in x 5.4 in)
Weight	0.84 kg (1.85 lb) receiver only, no radio model
	0.86 kg (1.90 lb) receiver only, radio model
	1.13 kg (2.49 lb) total weight including internal battery and UHF antenna
	3.96 kg (8.73 lb) items above plus range pole, Trimble TSC7 data collector and bracket

- **TEMPERATURE**

Operating	-40 °C to +65 °C (-40 °F to +149 °F)
Storage	-40 °C to +80 °C (-40 °F to +176 °F)

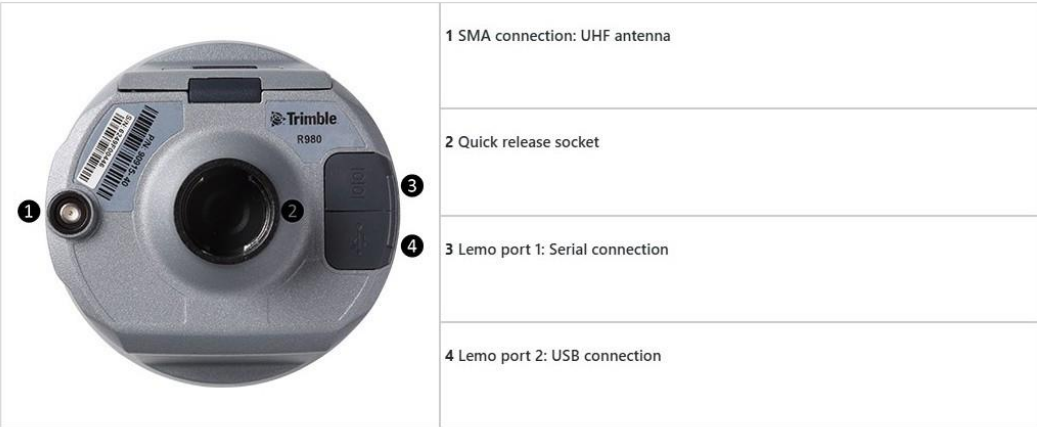
- **OPERATING TIMES ON INTERNAL BATTERY**

Rover	450 or 900 MHz receive	5.5–6.3 hours
	Cellular receive (Internal or Controller via Bluetooth)	7.0 hours
Base station	450 MHz transmit	3.7–5.8 hours (1.0 W transmit available only where legally permitted)
	900 MHz transmit (1.0 W)	6.0 hours (900 MHz transmit available only where legally permitted)
	Cellular transmit	7.0 hours

- **Communications and data storage**

Radio modem	Transmit power (Internal Radio)	0.1 W, 0.5 W, 1.0 W (1.0 W available only where legally permitted)
	Range	3-5 km typical with Internal Radio, 10 km optimal with External Radio
Cellular	Fully integrated, fully-sealed LTE compliant module with 2G/3G fallback	
Bluetooth	Fully-integrated, fully-sealed 2.4 GHz Bluetooth module	
Wi-Fi	Fully-integrated, fully-sealed 2.4 GHz Wi-Fi module	
Positioning rates	1 Hz, 2 Hz, 5 Hz, 10 Hz, and 20 Hz	
I/O ports	Serial, USB, TCP/IP, IBSS/NTRIP, Bluetooth	
Data storage	9 GB internal memory	





Receiver ports

Icon	Name	Connections
	Port 1	Device, computer, external radio, power in
	Port 2	Device, computer, USB flash memory stick, power in, power out
	RADIO	Radio communications antenna

CONTROLLER



8. Setting up the GNSS Equipment

8.1 GNSS Receiver Setup (Rover)

- Mount Trimble GNSS receiver R980 on survey pole.
- Insert internal batteries in Rover Receiver.
- Center and level using level bubble.
- Switch On Rover.
- Check tracking of multiple constellations (GPS, GLONASS, Galileo, BeiDou, NavIC if supported).
- Do e-bubble calibration & IMU alignment (**see Annexure-A**)

8.2 Controller Setup (Trimble Access)

- Power ON the controller.
- Launch **Trimble Access**.

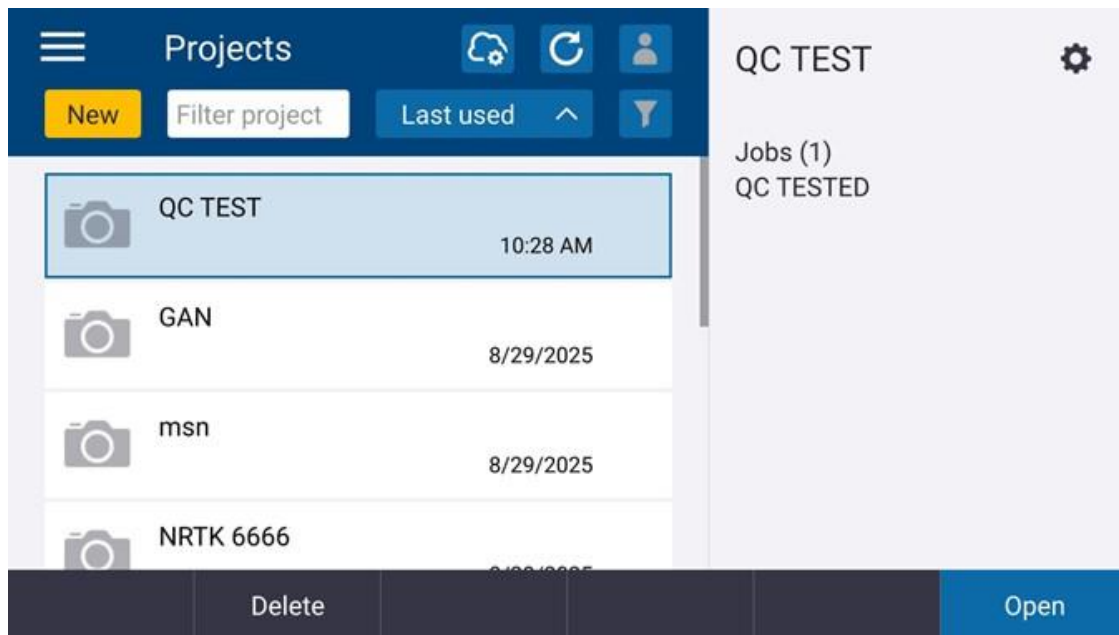
8.3 Internet Connection Setup

- Enable mobile data / Wi-Fi hotspot
- Verify stable internet connectivity
- Confirm sufficient signal strength (using Ping Test in Controller)

9. Procedure in Trimble Access

9.1 Create New Project & Job

- Open Trimble Access in Controller TSC5 or Getac, click on **New** to create a New Project



- Give Project Name e.g. " NRTK TESTING " and Click >>Enter>> **Create**

The screenshot shows the 'New project' form. The title is 'New project' with a back arrow. The 'Properties' tab is selected. The form has fields for Name, Description, Reference, Location, and Image. The 'Name' field contains 'NRTK TESTING'. The other fields have question marks. The bottom bar has 'Esc', a camera icon, and 'Enter' buttons.

Properties	
Name	NRTK TESTING
Description	?
Reference	?
Location	?
Image	?

- Inside project create job, Give a name to this new job e.g. " NRTK Test "

← New job: NRTK TESTING/NRTK Test ☆

☒ Create from template ☐ Create from JobXML or DC file

Job name: NRTK Test

Template: Metric Scale Only

Properties

Coord. sys.: Scale: 1.0000000000

Units (Dist.): Meters

Layer manager: None

Feature library: None

Esc Accept

9.2 Set Coordinate System, Units & CSV File Name

- Click on **Coord.sys.**
- Then Click on **Select from library**

Trimble Access

≡ Select coordinate system

Select coordinate system

☒ Scale factor only

☐ Select from library

☐ Key in parameters

☐ No projection / no datum

☐ Broadcast RTCM

Esc Next

Windows taskbar: Type here to search, Gold, 11:36, 13-10-2025

➤ In the **Select coordinate system**

- Select System: World wide/UTM from drop down menu
- Select Zone: Calculate and select from dropdown menu as per your working area Longitude (44N for Hyderabad)
- Select Local datum: WGS 1984 (7P) from drop down options
- Global reference datum: WGS 1984 (Default)
- Global reference epoch: 2005(Default)
- Displacement model: ITRF2020 (Default)
- Tectonic plate: Detect automatically
- Use Geoid model Click YES (if any local Geoid model is available) & select appropriate GGF file or else select NO.

Trimble Access

Select coordinate system

System
World wide/UTM

Zone
44 North

Local datum
WGS 1984 (7P)

Global reference datum
WGS 1984

Global reference epoch
2005.00

Displacement model
ITRF2020

Tectonic plate
(Detect automatically)

Use geoid model
☐ No

Use datum grid
No

Coordinates
Grid

Project height
500.000m

Esc Point Key in Store

Type here to search

11:37
13-10-2025

- Use datum grid: No
 - Coordinates: Grid.
 - Project Height: Give average Height of the working area.
- Click >>**Store**

- Now click on **Units(Dist)**

Trimble Access

← Job properties: RTK RADIO ☆

Job path ...\\Projects\\NRTK TESTING\\NRTK Test.Job

Properties

Coord. sys.	44 North (World wide/UTM)
Units (Dist.)	Meters
Layer manager	None
Feature library	None
Cogo settings	Ground
Additional settings	Off
Media file	Previous point
Reference	?
Description	?
Operator	?

Esc Accept

Type here to search

Indi... ENG 11:37 13-10-2025

- Select options from following window as per convenience and Click on **Accept**

Trimble Access

≡ Units

Display

Distance display	0.001	Coordinate display	0.001
Area display	0.001	Volume display	0.001
Mass display	0.001	Angle display	1"
Lat / Long	DDD.MMSS	Coordinate order	East-North-Elev
Station display	1+000.0	Laser VA display	Inclination
Time format	Local date/time	Precision display	DRMS

Esc Accept

Type here to search

Indi... ENG 11:38 13-10-2025

- Now again Click>> **Additional settings**

Additional settings

Feature library

Use attributes of base code
☒

Add to CSV file

Enable
☒ Yes

CSV file name
NRTK Testing

Point name range for job

Apply point name range
☐ No

Esc Enter

- **Enable** Add to csv & give name e.g. " NRTK Testing " to .csv file. Click >> **Enter**

Trimble Access

Job properties: RTK RADIO

Job path: ...\\Projects\\NRTK TESTING\\NRTK Test.job

Properties

Coord. sys.	44 North (World wide/UTM)
Units (Dist.)	Meters
Layer manager	None
Feature library	None
Cogo settings	Ground
Additional settings	CSV file
Media file	Previous point
Reference	?
Description	?
Operator	?

Esc Accept

Type here to search

Air...

ENG 11:39 13-10-2025

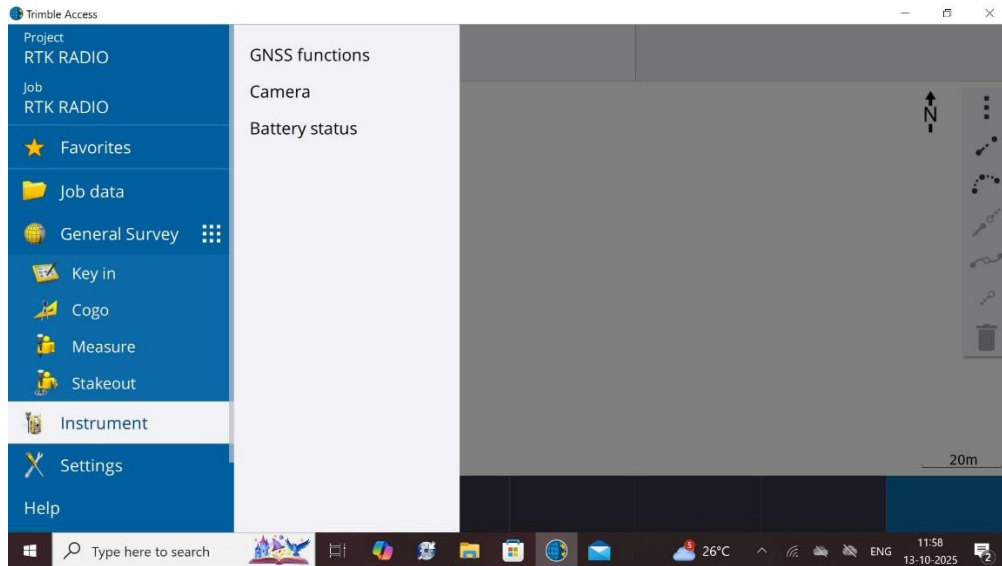
- Click >> **Accept**

Now a New Project & a New Job with a defined projection system has been created.

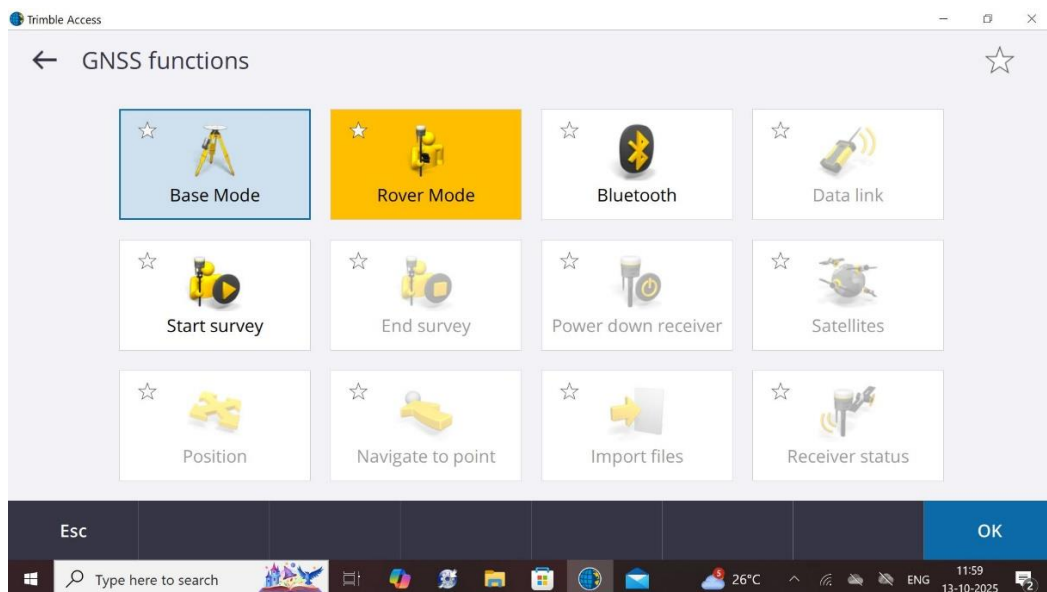
10. Configure Rover Receiver with Controller

In order to configure Rover Receiver with Controller

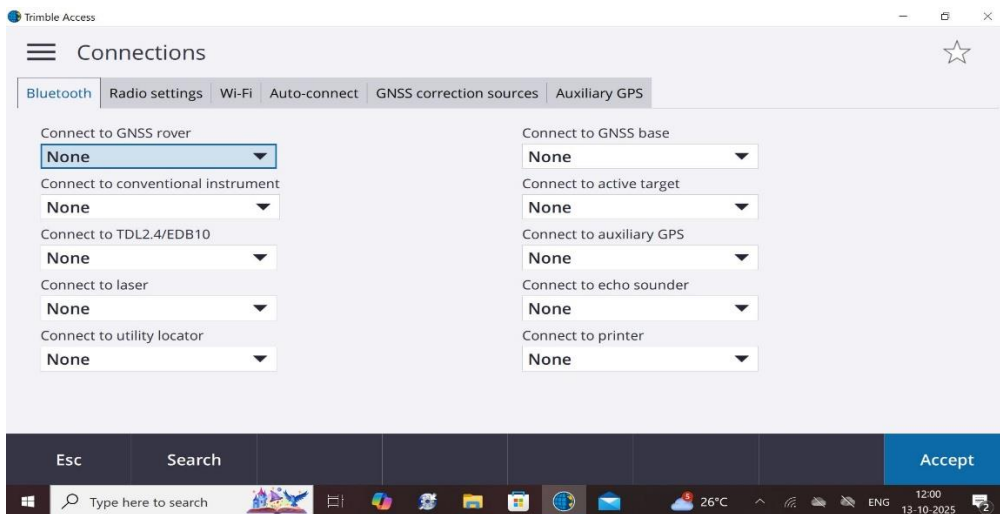
- Click on **Instruments >> GNSS Functions**



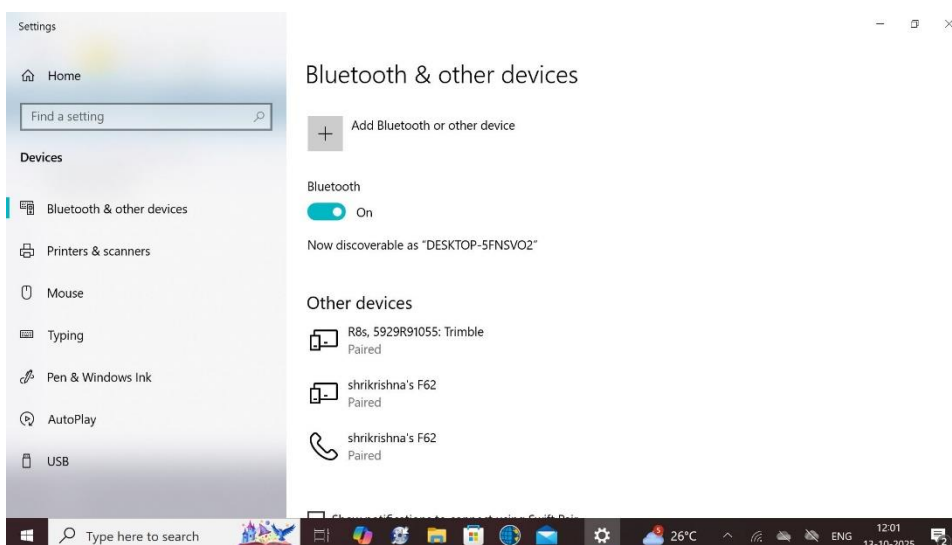
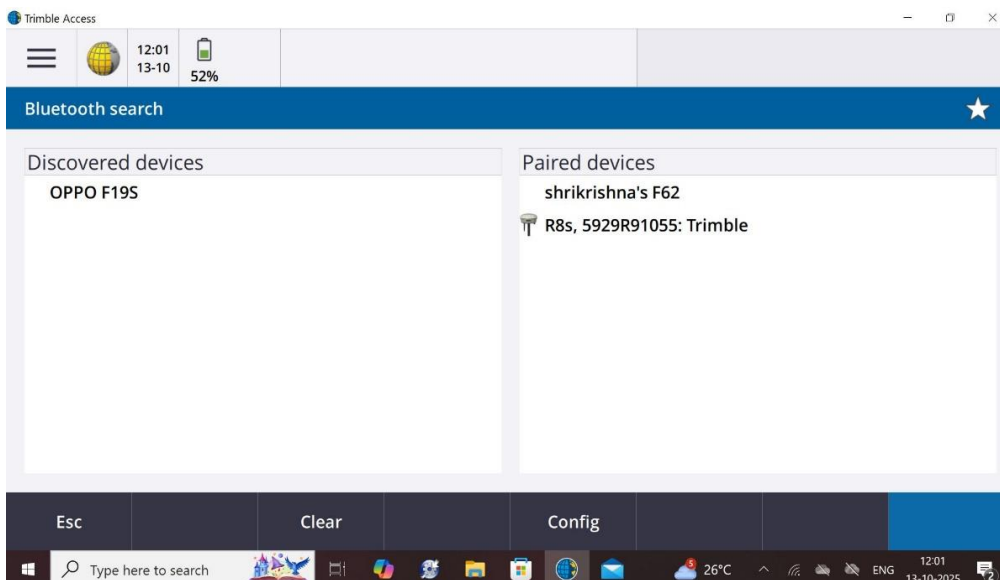
- Click on **GNSS Functions>> Bluetooth**

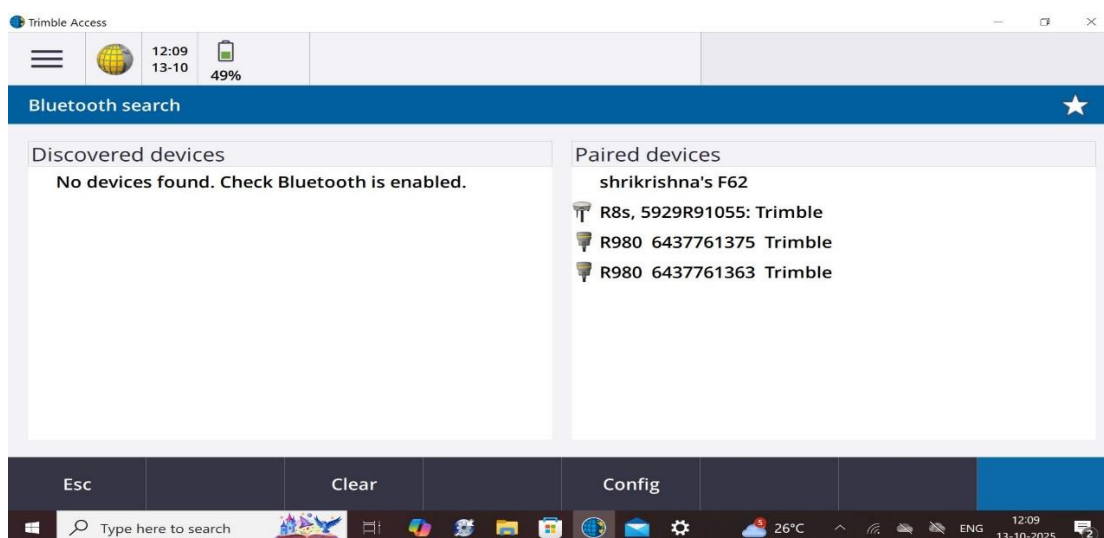
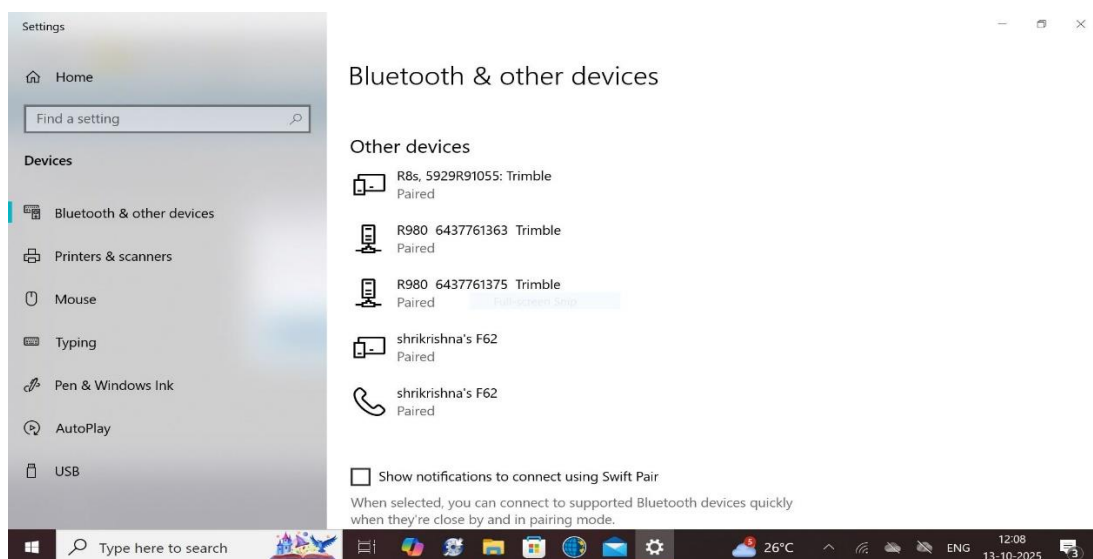
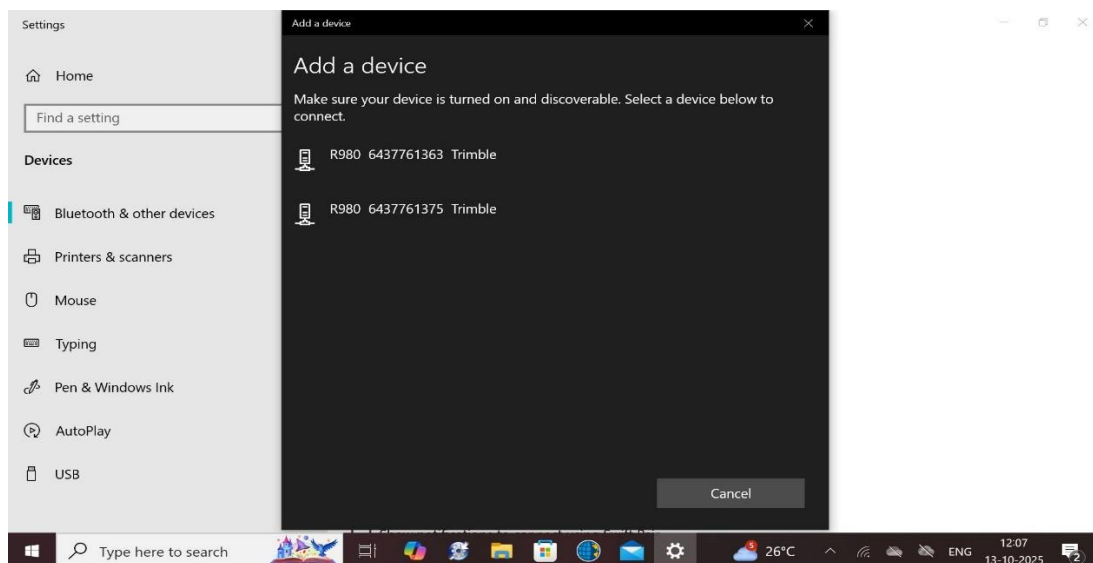


➤ Now Click on **Search**

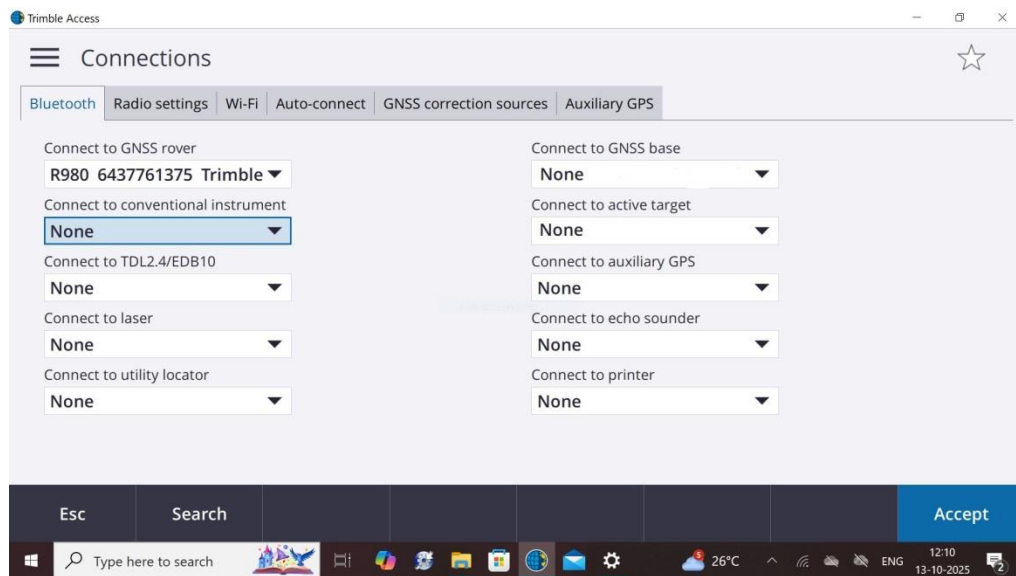


➤ Click on **Config** and Pair Base & Rover Receiver as per following steps:





- Select respective GNSS Rover from Paired Receivers as follows:



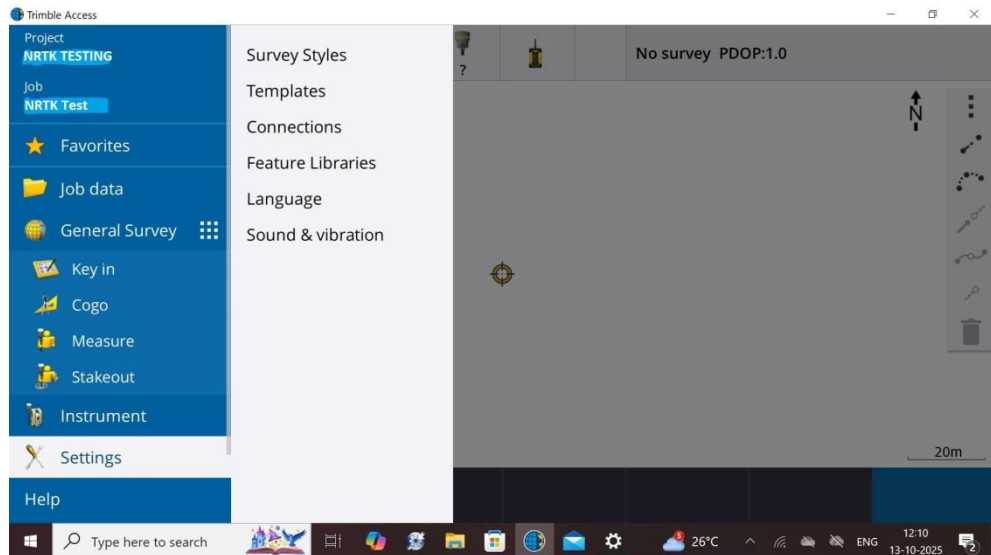
11. Creating NRTK Survey Styles

In order to create a NRTK Survey Style in NRTK Method we have to define following things:-

- Rover Option
- Rover Data Link

Following are the steps mentioned to complete above settings.

- Create a New Survey style for NRTK method, for that click main menu bars
- Go to **Settings >> Survey Styles**



- Click on **New** and Give name of style e.g. **NRTK Testing** and **Accept**.

Survey Styles

Name	Modified
666	8/28/2025
msn	8/28/2025
peera	8/28/2025
RTK	3/23/2025
test	4/22/2025

Esc New Copy Delete Edit

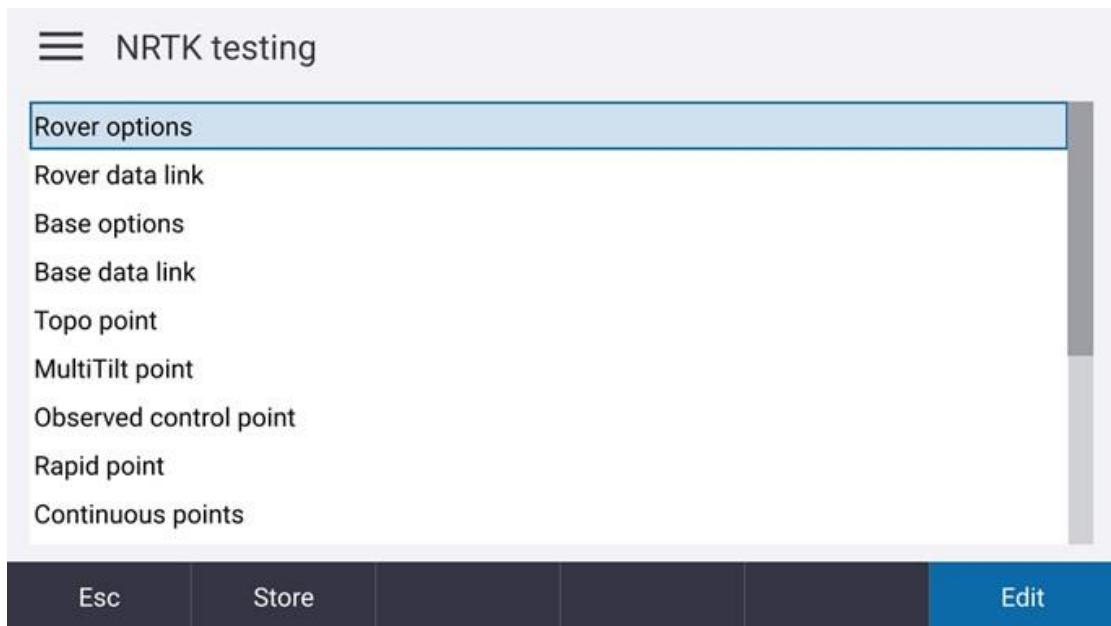
Style details

Style name: NRTK testing

Style type: GNSS

Esc Enter

A new window will open as follows:



- IN Rover Options click on **Edit**

In Rover Options

- Select Survey Type as RTK
- Broadcast Format: RTCM(VRS)
- Provide specifications of GNSS Antenna using.
- Check on IMU tilt compensation & eBubble Functions if desirable.
- Keep Satellite Differential Off.
- Enter Elevation Mask angle, PDOP value.
- Check on all GNSS Constellation.
- Enter Roving precision values in Horizontal & Vertical as per requirements.
- Check on xFill if it is desirable.

Rover options

Survey type: **RTK** Broadcast format: **VRS (RTCM)**

Antenna

Type: **R980 Internal** Measured to: **Bottom of quick release**

Antenna height: **2** Part number: **128221-XX**

Serial number: **?**

Tilt

Esc Enter

Rover options

Use station index: **Any**

Prompt station index: ☒

Elevation mask: **10°**

PDOP mask: **6.0**

Satellite differential: **Off**

GNSS Signal Tracking

GPS: ☒ Use L2e: **Yes**

GPS L2C: ☒ L5: ☒

GLONASS: ☒ Galileo: ☒

Esc Enter

Rover options

GNSS

☒ GPS ☒ Use L2e

☒ QZSS ☒ BeiDou

☒ NavIC ☒

Roving precision

Auto tolerance: ☒ No

Horizontal tolerance (DRMS): **0.020m** Vertical tolerance (1 sigma): **0.040m**

xFill: ☐

Esc Accept

- Click on **Accept**.
- After completing above step a new window will appear **Rover Data Link**

☰ NRTK testing

- Rover options
- Rover data link
- Topo point
- MultiTilt point
- Observed control point
- Rapid point
- Continuous points
- Stakeout
- Site calibration

Esc Store Edit

- Click on **Edit**
- Following window **Edit GNSS correction source** will appear in which we have to define new GNSS correction source via which Rover would be connected to CORS Server. Here we are giving this source name e.g. **NRTK Testing**

☰ Edit GNSS correction source

GNSS correction source name

NRTK Testing

NTRIP Configuration

Use RTX (internet) ☐ No

Use NTRIP ☒ Yes

Use NTRIP v1.0 ☐ No

Use proxy server ☐ No

Connect directly to Mountpoint ☐ No

NTRIP username

NIGST3

NTRIP password

.....

Esc Enter

- In above window define NTRIP Configuration by clicking **Use NTRIP >> Yes** & keep remaining all off. Also provide
 - NTRIP Username: *****
 - NTRIP Password: *****

Edit GNSS correction source

Use NTRIP v1.0 ☐ No

Connect directly to Mountpoint ☐ No

NTRIP password

IP address

Send user identity info ☐ No

Use proxy server ☐ No

NTRIP username

IP port

Use TLS encryption ☐ No

Esc Enter

- Provide IP ADDRESS for Region in which you are working.
 - For **Region 1** IP Address is **103.205.244.106** & IP port **2101**
 - For **Region 2** IP Address is **103.206.29.4** & IP port **2105**
 - Send user Identity info: NO
 - Use TLS encryption: NO
- Click >> **Enter**
- In next window Click on **GNSS Internet source** and select **Controller Internet** also provide Internet connection to Controller via SIM/Wi-Fi.

Rover data link

Type
Internet connection

GNSS correction source

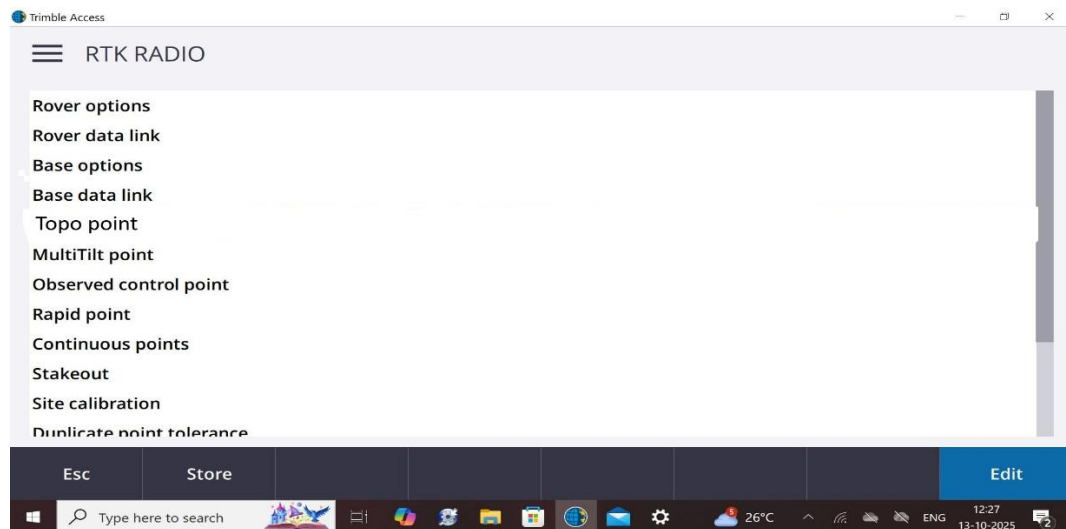
GNSS internet source

Prompt for GNSS correction source ☐

Prompt for internet source ☐

Esc Enter

- Click on **Accept**.



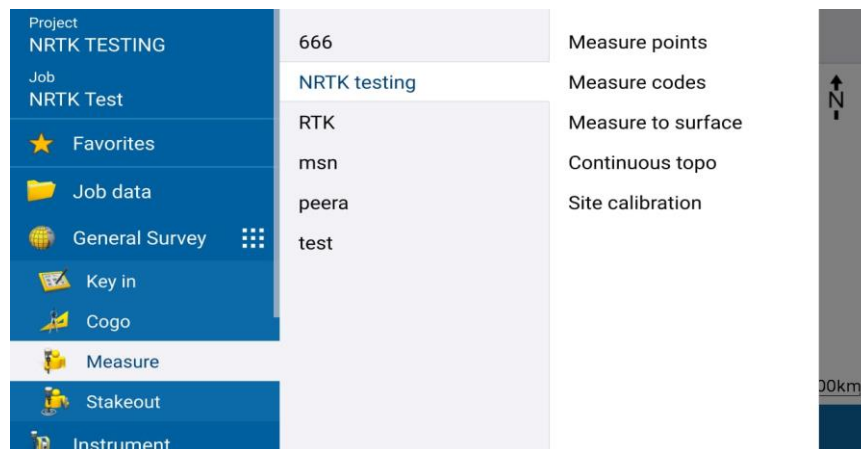
- Click >> **Store**. With this new survey style for NRTK has been created.

Using above steps our Rover Receiver is connected with Controller. Now in above window we can see other options also in which **Topo point**, **MultiTilt point**, **Observed Control point** & **Rapid point** are the methods of observing a point with different accuracies. (see **Annexure B**). Above methods can be selected and used as per the survey needs.

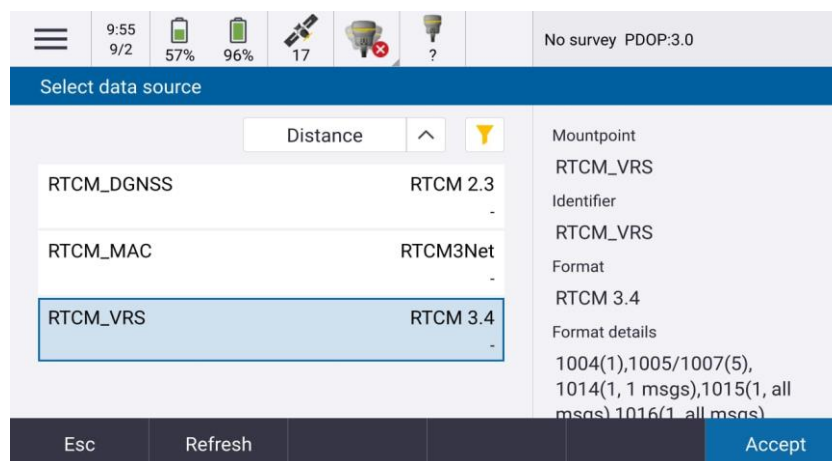
Before starting the survey the controller should be connected to Internet via wifi , modem or cable, active internet connection is must for NRTK method.

12. Field Procedure for NRTK

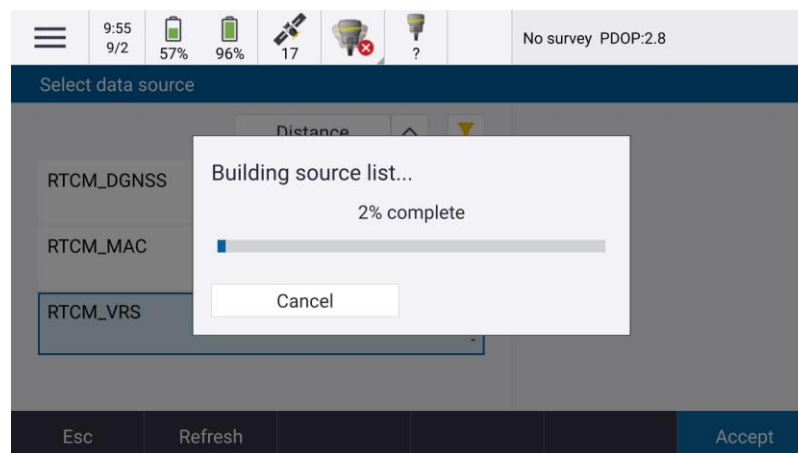
After the completion of above mentioned steps now Click on **Menu Button**>>
Measure >> NRTK Testing (Survey style as created in above steps) >> **Measure Points**



- It will prompt for selecting the data source, RTCM_VRS (whichever Broadcast format you have given during the creation of style)



- Click >> **Accept**



At the end of above step the controller will be connected to the CORS server to provide corrected data in real time. Following screen would be visible.

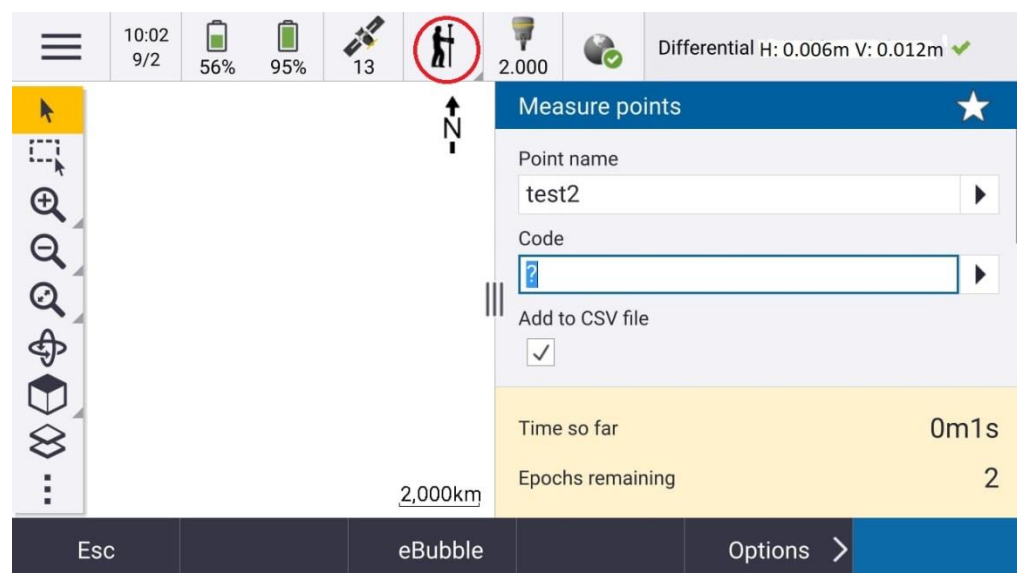
➤ In the above pictures

On the north top

- The vertical 3 lines symbol is for entering main menu
- Next Time & date
- Battery status of Controller
- Battery status of Receiver
- Number of Satellites being tracked at that time (by clicking you can see details of satellites being tracked).
- Next option is Tilt compensation(IMU) The Trimble R980 has built-in feature of Tilt compensation. If a red cross is shown with Receiver icon it means that Receiver is not IMU calibrated, so in order to do IMU calibration rotate the pole mounted with receiver without lifting the tip from ground afterwards

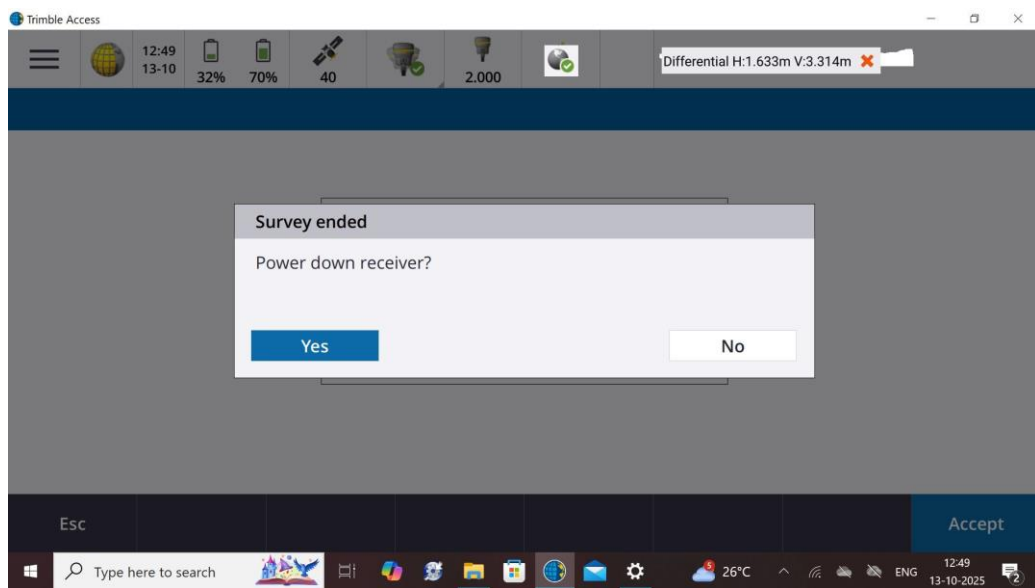
red cross will change to Green tick , it means that receiver is now IMU calibrated or ready to take observations in tilt position.

- Next option denotes the Pole height it is fixed 2m for NRTK survey.
 - Next Globe symbol with green tick represents connection with the server, if connection breaks for any reason like internet connection disturbance it will turn to Red Cross with globe symbol. Survey is possible only after the connection is restored.
 - Next icon shows the horizontal and vertical precision values, if these are under given tolerances green tick will come after that only point will be recorded.
 - E-bubble function, is also available in Trimble R980 before using this function it needs to be calibrated. **(See Annexure A)**
 - In Measure point option give point name to be observed.
 - Next if required you can give any code
 - Next in .csv file see that concerned csv file name is selected
 - Measure as **Topo point**. (Measuring method see **Annexure B**)
 - Select Measured to **Bottom of Quick Release** (Height measuring method)
 - Click on **Options** icon and select **QC1:&QC2** parameters and increase time of observation if required.
- Then Click >>**Enter** >> **Measure**. Here You can observe during survey the above icon shown in red circle will change.



- Click >> **Store**
- Now to see observed points Click on Menu button, then Click >> **Favorites** >> **Point manager**. A list would be opened with all observed points.
- After completion of observations Click on Menu button >> **Measure** >> **End GNSS**

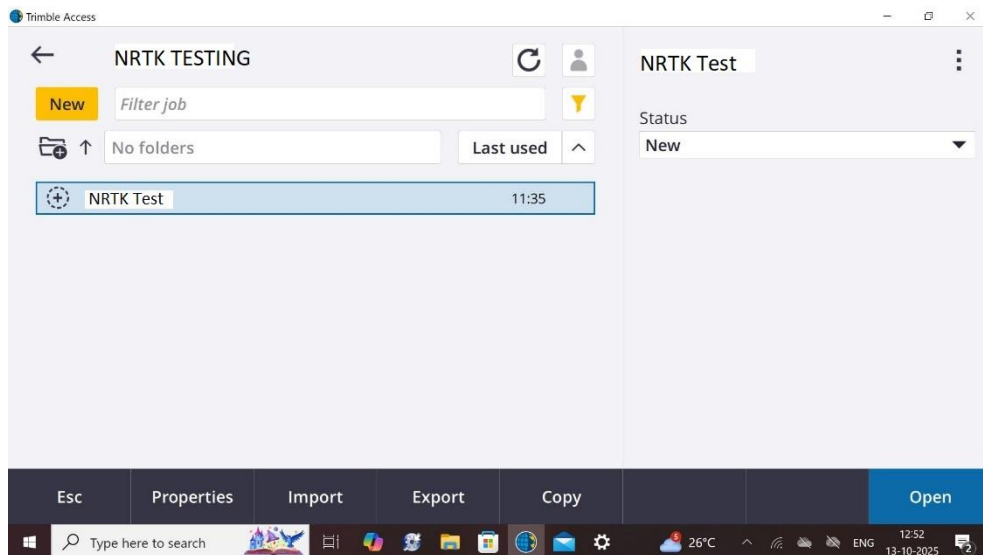
survey to stop observations. Following window will appear.



- To Power down receiver Click >> **Yes**

13. Downloading the Data

- After Ending the survey, Go to job data in menu of Trimble Access and Open the Project
Select the Job **NRTK Test** under Project **NRTK TESTING**
At the bottom you will find **Export** option >> Click



- Here in Export window you can select your desired format for observed data, also you can give path where exported data would be stored. Here we are selecting *.CSV format for our observed data. Click >> **Accept**

Export

File format

Comma Delimited (*.CSV, *.TXT)

File name

NRTK Test.csv

Point name

Field1

Point code

Field5

Northing

Field3

Easting

Field2

Elevation

Field4

Field delimiter

<comma>

View created file

☒

Esc

Accept

➤ Click on **All points**

Trimble Access

12:52
13-10

30%

Select points

Enter single point name

Select from list

All grid points

All keyed in points

Points within radius

All points

Points with same code

Points by name range

Section of job

Map selection

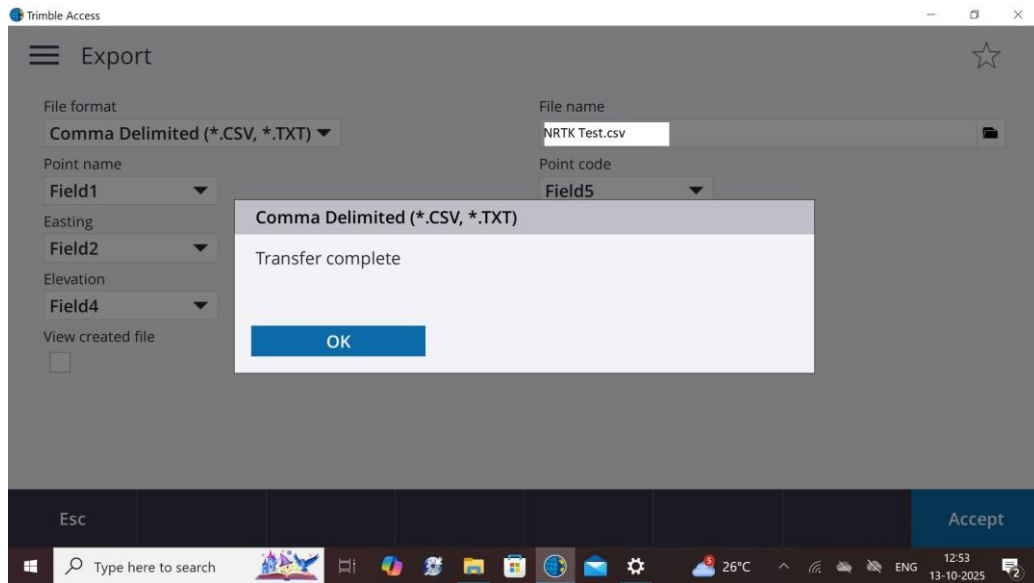
Esc

Accept

Type here to search

26°C

12:52
13-10-2025



➤ Click >> **OK**

14. Conclusion

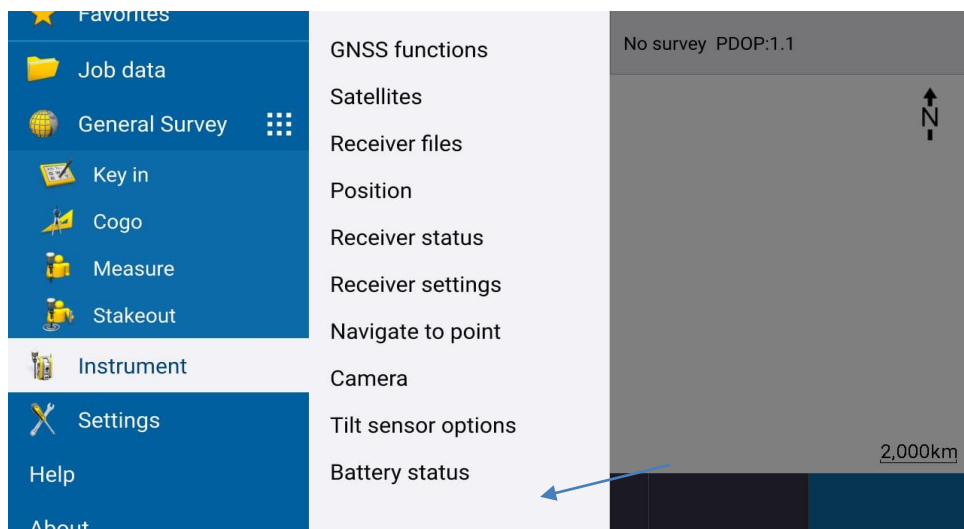
This SOP establishes a standardized and reliable procedure for conducting NRTK surveys using the Trimble R980 GNSS receiver. Adherence to the defined setup, operational steps, and quality control measures ensures accurate, consistent, and efficient positioning results. Proper implementation of this SOP enhances field productivity, data reliability, and overall survey confidence.

15. Annexure A : E-bubble calibration

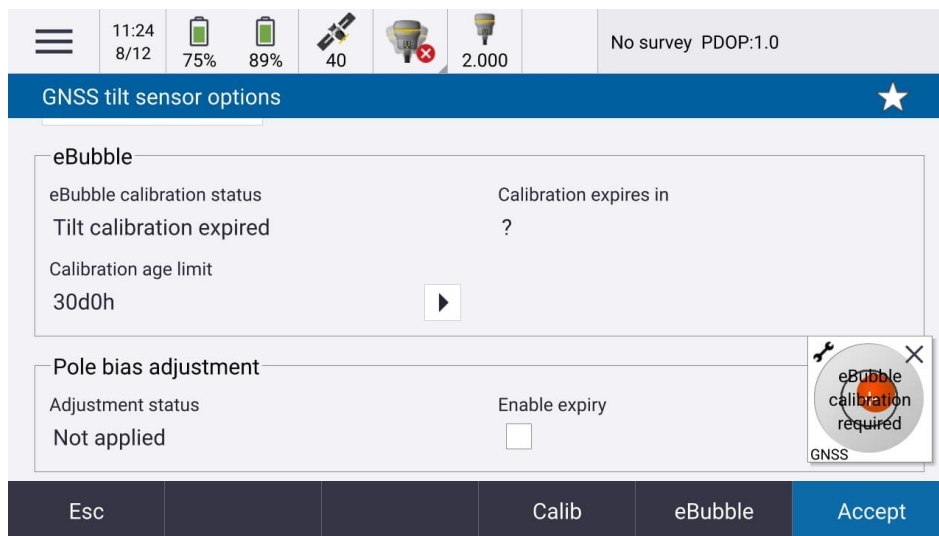
E-bubble calibration

The following steps will guide you in performing the e- bubble calibration.

1. Set up the Receiver
 - Connect R980 to controller via Bluetooth.
 - Mount the receiver on a bipod.
 - Level it using the bubble level.
 - Confirm stability before proceeding.
2. Make sure you properly levelled the receiver on bipod.
3. Open any created Job. Click on Menu Button >> **Instrument** >> **Tilt sensor options**



4. In **GNSS Tilt sensor options** Click on **Calib** button at bottom.



The screenshot shows the 'GNSS tilt sensor options' screen. At the top, a status bar displays icons for battery (75%, 89%), signal strength (40), and other sensors. Below the status bar, the screen is divided into two main sections. The first section, titled 'eBubble', shows 'eBubble calibration status' as 'Tilt calibration expired' and 'Calibration expires in' as '?'. It also shows 'Calibration age limit' as '30d0h'. The second section, titled 'Pole bias adjustment', shows 'Adjustment status' as 'Not applied' and 'Enable expiry' as an unchecked checkbox. A circular warning icon with a red center and the text 'eBubble calibration required' is overlaid on the right side of the screen. At the bottom, there are five buttons: 'Esc', 'Calib', 'eBubble', and 'Accept'.

GNSS tilt sensor options

eBubble

eBubble calibration status
Tilt calibration expired

Calibration expires in
?

Calibration age limit
30d0h

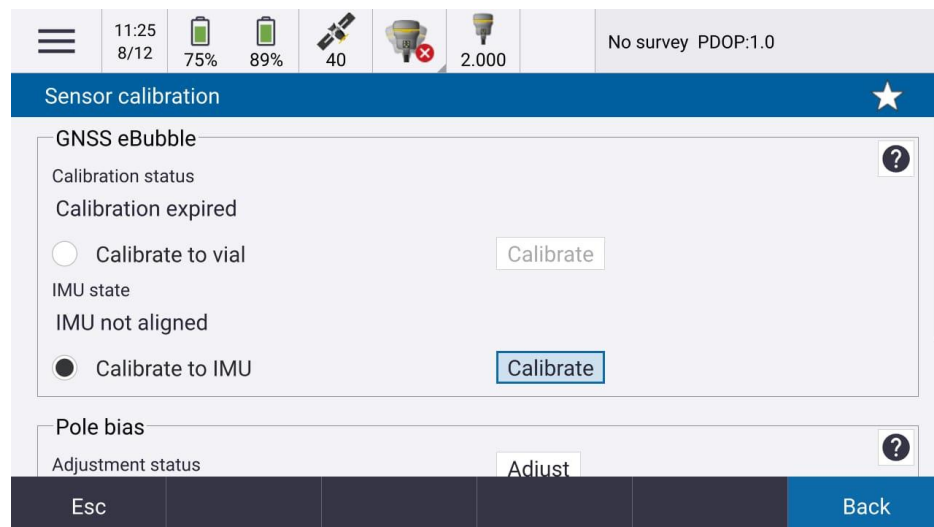
Pole bias adjustment

Adjustment status
Not applied

Enable expiry
☐

Esc Calib eBubble Accept

5. Now in upcoming window **Sensor calibration** select **Calibrate to IMU** and Click on **Calibrate** button.



The screenshot shows the 'Sensor calibration' screen. At the top, a status bar displays icons for battery (75%, 89%), signal strength (40), and other sensors. Below the status bar, the screen is divided into two main sections. The first section, titled 'GNSS eBubble', shows 'Calibration status' as 'Calibration expired'. It has two radio buttons: 'Calibrate to vial' (unselected) and 'Calibrate to IMU' (selected). There are 'Calibrate' buttons next to each radio button. The second section, titled 'Pole bias', shows 'Adjustment status' as 'Adjust'. At the bottom, there are five buttons: 'Esc', 'Back', and three unlabeled buttons.

Sensor calibration

GNSS eBubble

Calibration status
Calibration expired

☐ Calibrate to vial

☒ Calibrate to IMU

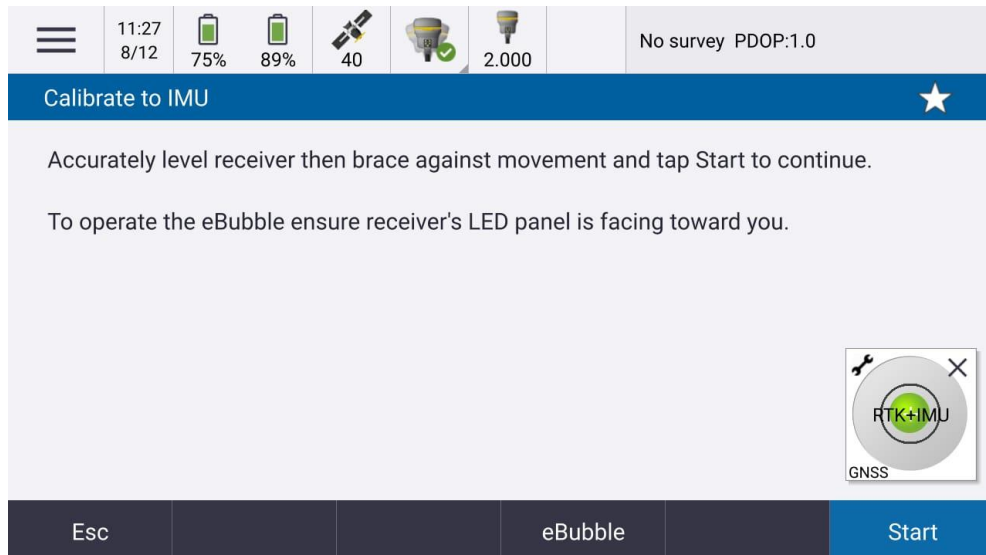
IMU state
IMU not aligned

Pole bias

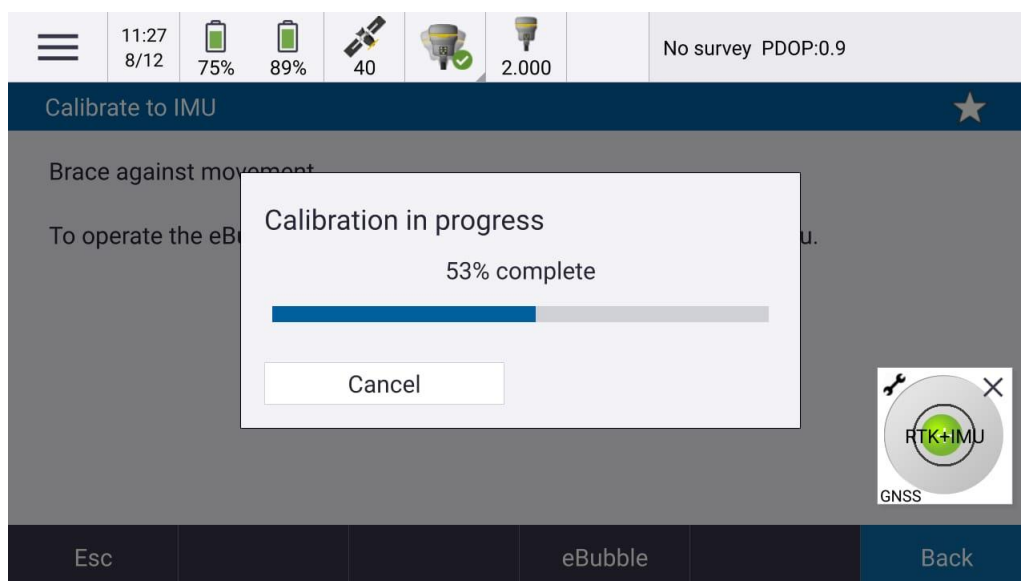
Adjustment status
Adjust

Esc Back

6. Ensure the receiver is properly leveled and stand with controller facing the LED panel of the receiver and Click on **Start** button.



7. Next you can see the calibration in progress



8. After completion of calibration e- bubble level will match to the level on the receiver connected to the bipod and calibration process is completed.
9. E-bubble calibration has to be observed periodically and must be performed if required.

Note:- This maneuver should be done only when Controller is well connected to Receiver and observations are about to be start (RTK+IMU initialization).

IMU Alignment/Calibration

In order to IMU Alignment of Receiver, after complete setup of Receiver & Controller and just before the commencement of observations, Receiver mounted on Survey pole should be moved in round motion while keeping the tip of Survey pole tip stable on ground.

16. Annexure B : Various Positioning methods

Various Positioning methods in RTK/NRTK

In Trimble Access application following positioning methods are provided to **Measure Point**:-

- Topo Point
- Rapid Point
- Multi Tilt Point
- Observed Control Point

Topo Point

A **Topo Point** is a point measured to **capture ground features** and terrain details during a **topographic survey**.

Rapid Point

A **Rapid Point** is a point observed with **short-duration repeated measurements** to obtain **better accuracy than a topo point**, but **without the full rigor of a control point**. It is commonly used in **RTK surveys** when you want **extra confidence** but **faster work**.

Multi Tilt Point

A **Multi-Tilt Point** is a point measured **several times at different pole tilt angles** using a **tilt-compensated GNSS receiver** (like Trimble R980), to **verify and improve confidence** in the measured position.

Observed Control Point

An **Observed Control Point** is a **high-accuracy reference point** whose coordinates are **established by observation** (GNSS static, fast static, or repeated RTK).

Comparison

Point Type	Method	Accuracy	Use
Topo Point	Single vertical	cm	Mapping Grade
Rapid Point	Repeated (time-based)	cm (better)	Temporary control
Multi-Tilt Point	Repeated (tilt-based)	cm (high confidence)	Validation / checks
Observed Control Point	Long static / repeat	mm–cm	Primary control (Survey Grade)

Continuous Topo

Continuous Topo is a feature-mapping method in Trimble Access that allows you to collect many topo points automatically while moving, instead of storing points one-by-one.

It continuously records positions at a set time or distance interval while the rover is moving with RTK fixed solution.

Ideal for linear features like Road, Railway, Tank Bund etc.

For getting observations in Trimble Access in ANY method whether it is RTK,NRTK OR PPP (RTX), Click on **Menu** Button >> **Measure** >> **RTK RADIO** (created survey style) >> **Continuous Topo** . In upcoming window we have to fill the starting **Point Id** and select **Method** (e.g. Fixed time etc. and enter that time interval in which you want the points) AND rest settings are same like in **Measure point** .

Now **Click** on **Start** tab in Right Corner in bottom, after this you will get points in a series within given time interval. To see these points you may go to **Point Manager**.

17. Annexure C : Stakeout

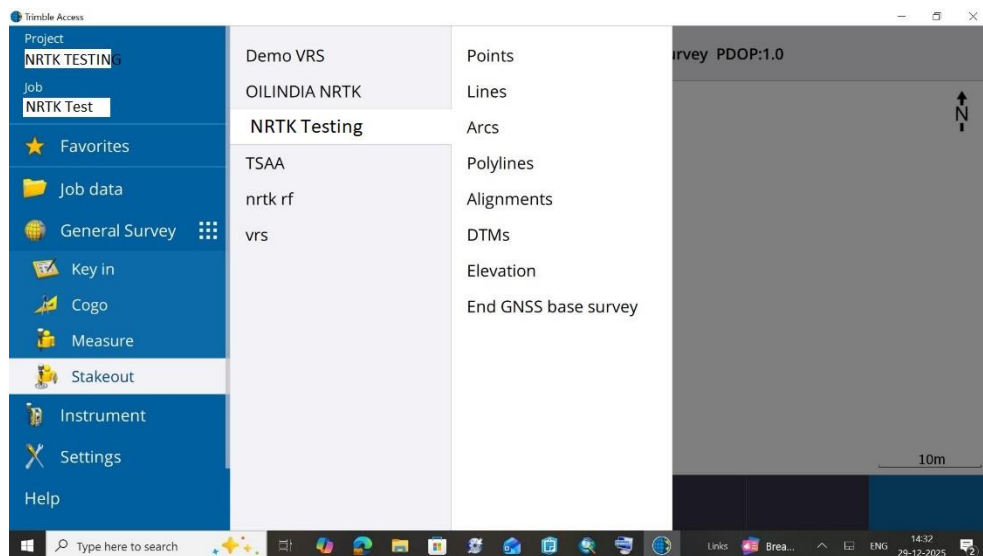
Stakeout

Stakeout is the process of **setting out (marking) design points on the ground** using GNSS guidance.

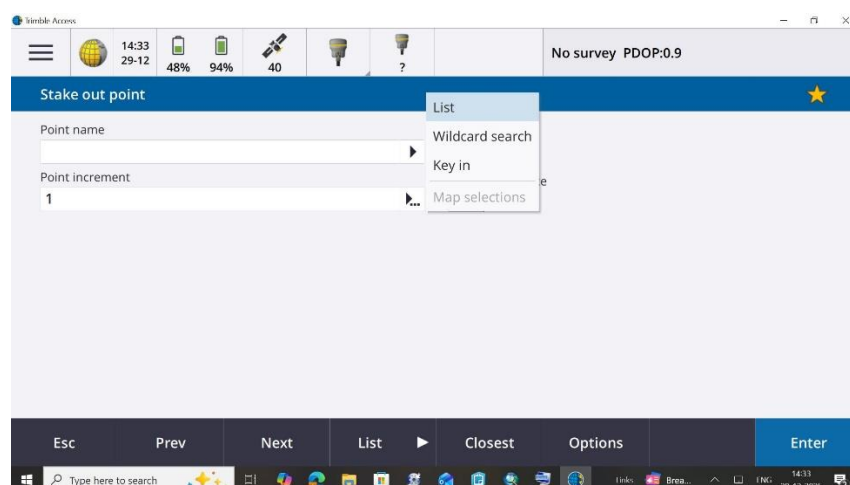
Purpose of Stakeout is to transfer design co-ordinates on ground. It is useful in Pillar/ Boundary points relocation.

Following is the process to do Stakeout in Trimble Access:

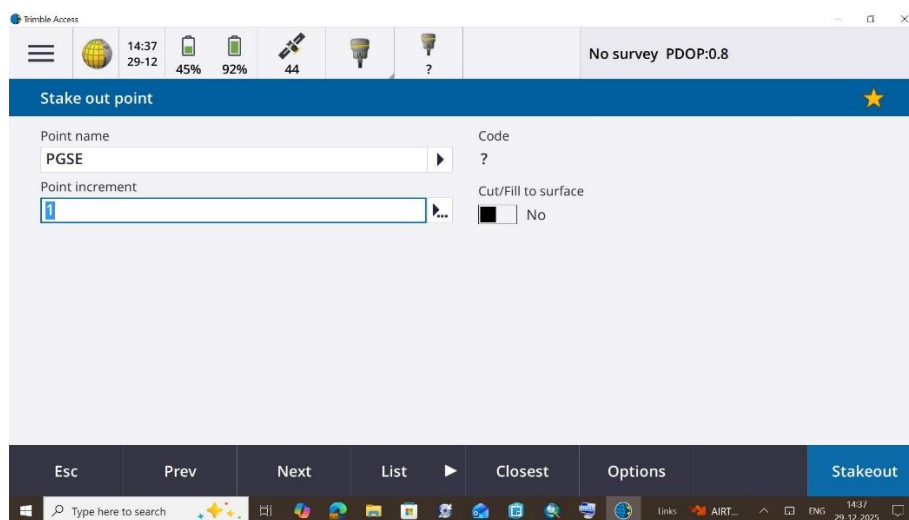
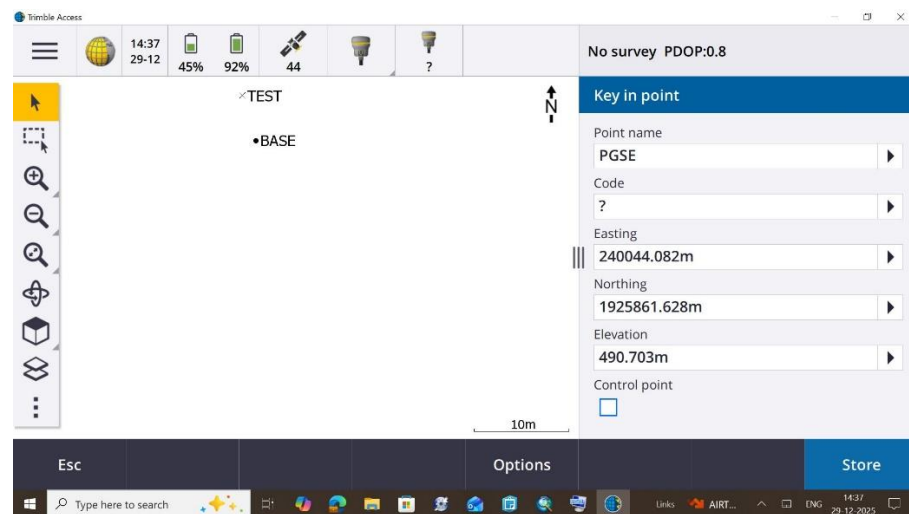
- Click >> **Menu >> Stakeout >> NRTK Testing (Created style) >> Points**



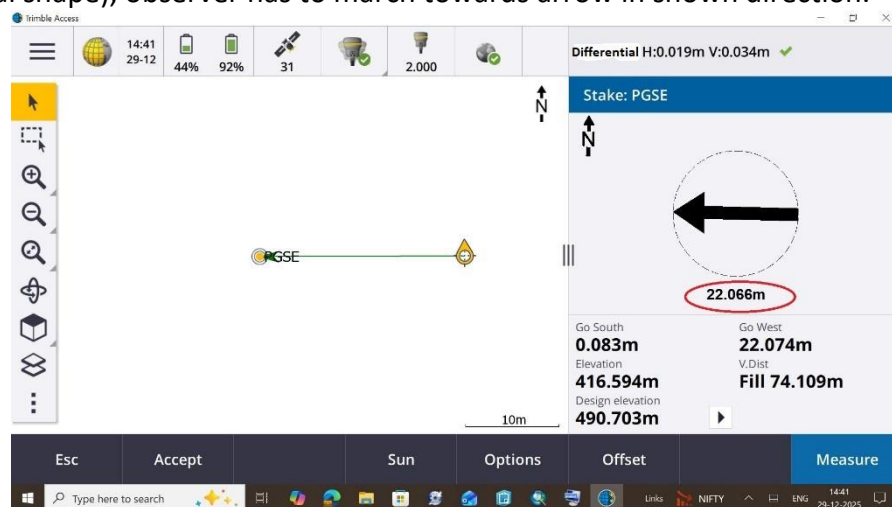
- In upcoming window, select Point to be relocate on ground, either by already installed **List** in the controller or **Key in** the co-ordinates of point to be relocate.



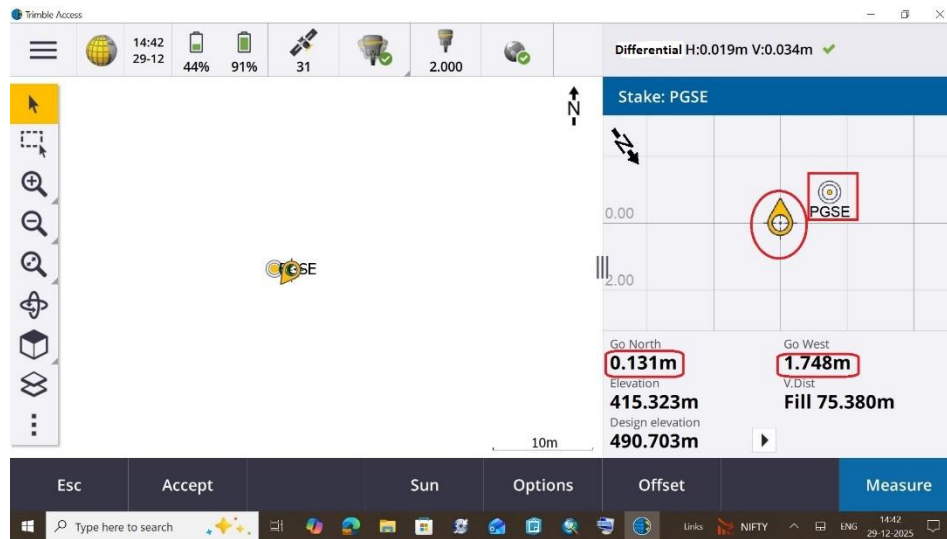
- Insert the co-ordinates Click >> **Store** >> **Stakeout**



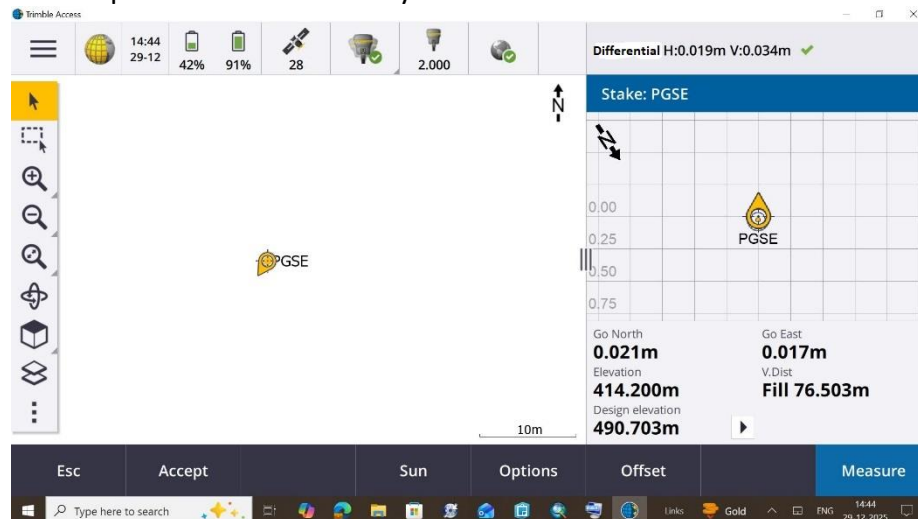
- In upcoming window distance between desired point & Rover is shown (in marked Red oval shape), observer has to march towards arrow in shown direction.



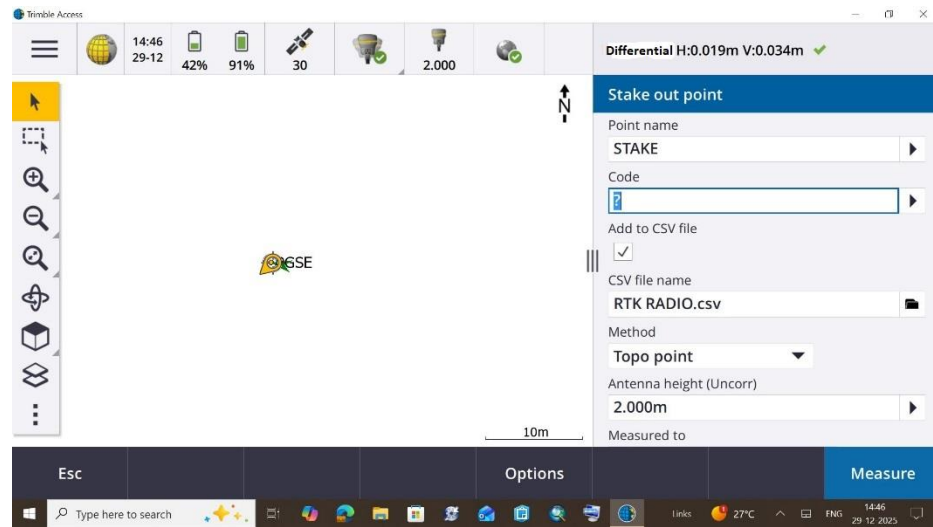
- When Rover comes near to desired point , a grid window appears wherein desired point (In marked Red color oval shape) and Rover (In marked square shape) are shown, also the distance in Easting and Northing also appears below the Grid box as follows:



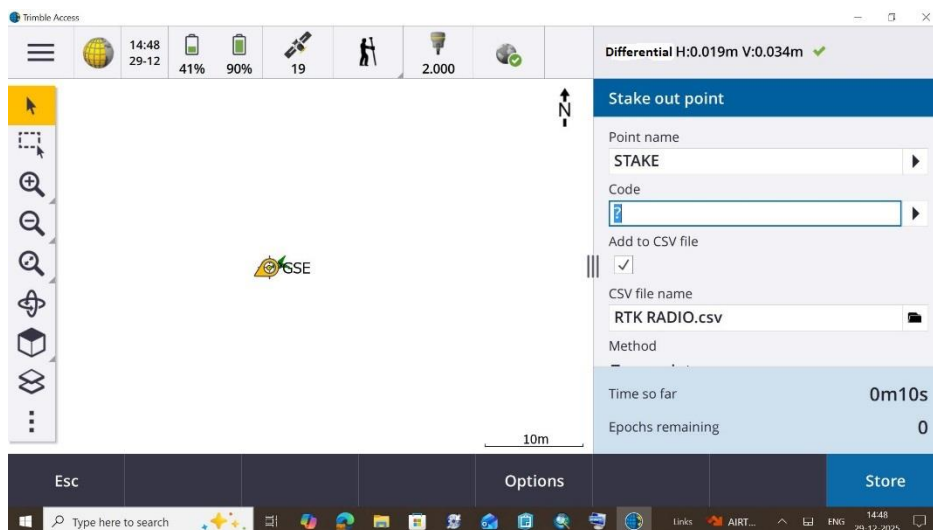
- When Desired position is coincided by Rover then Click >> **Measure**



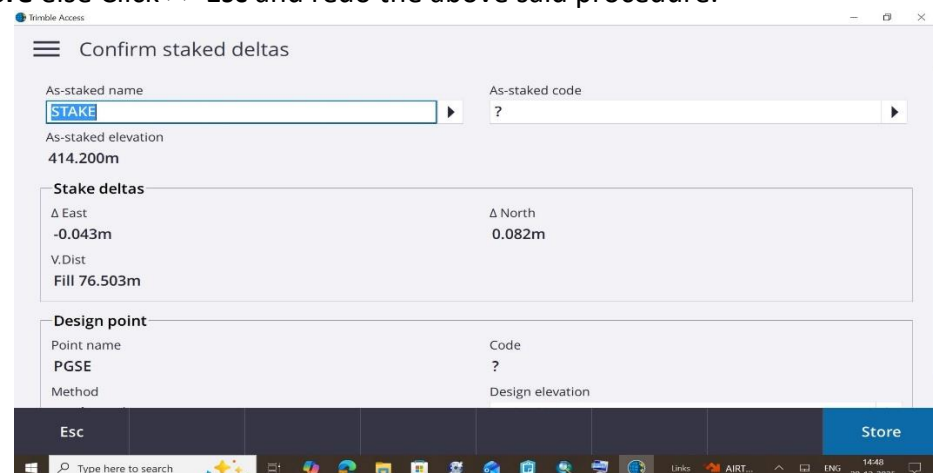
- In next window enter name of **Stake out point** name e.g. STAKE then Click >> **Measure**



- Click >> **Store**



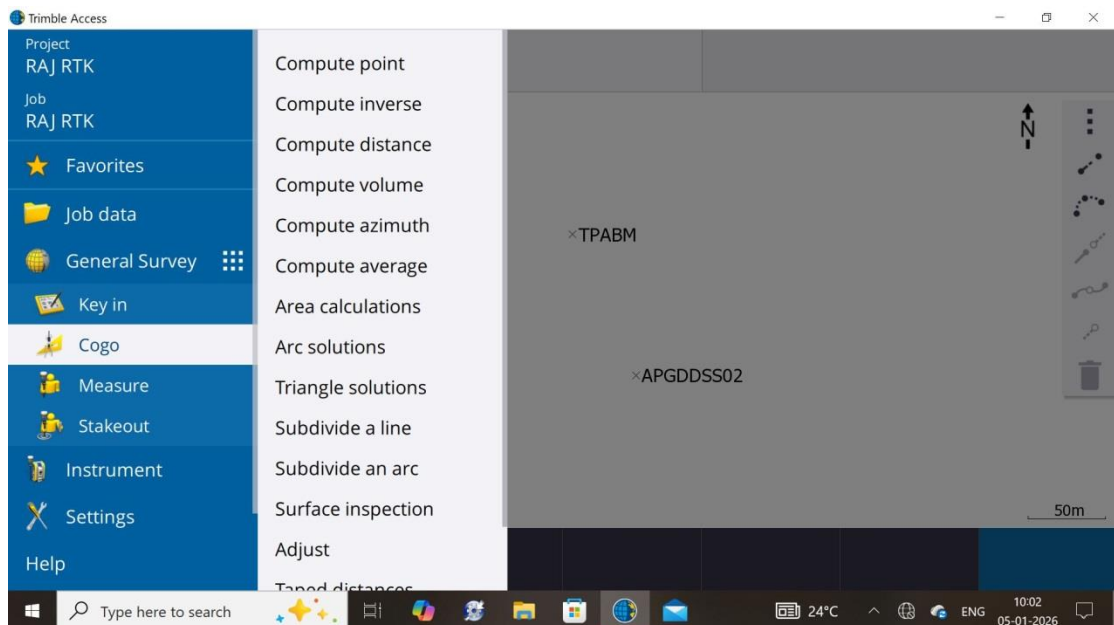
- In last step Stake out Deltas are shown, if they fulfill the stakeout criterion then Click >> **Store** else Click >> **Esc** and redo the above said procedure.



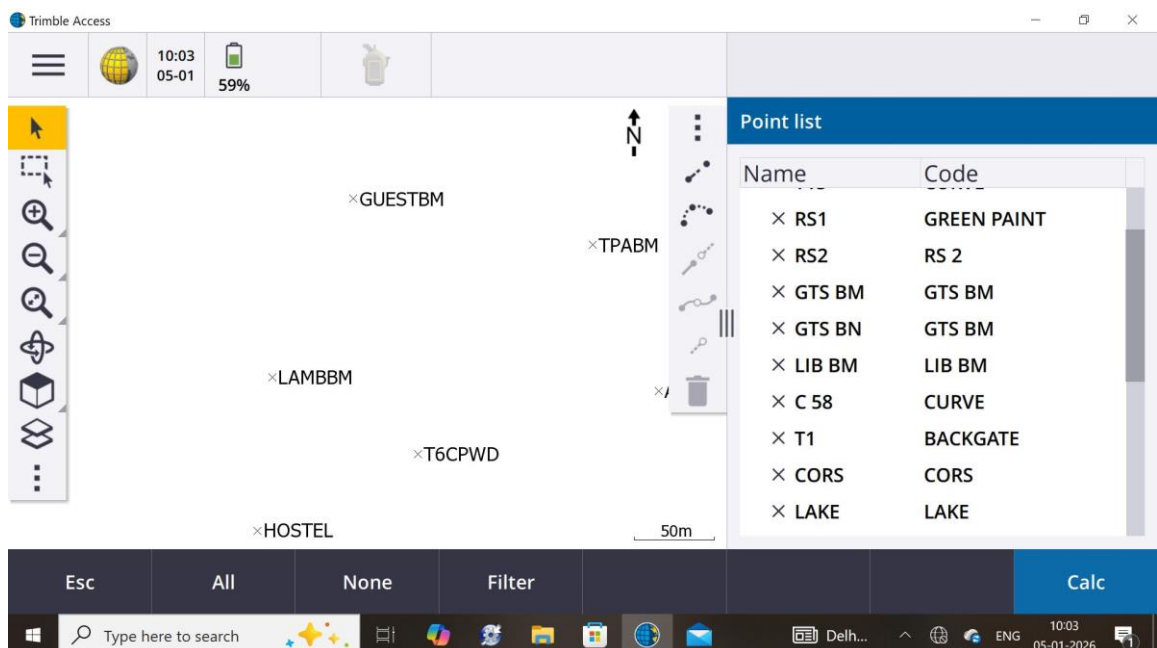
18. Annexure D : Area Calculation

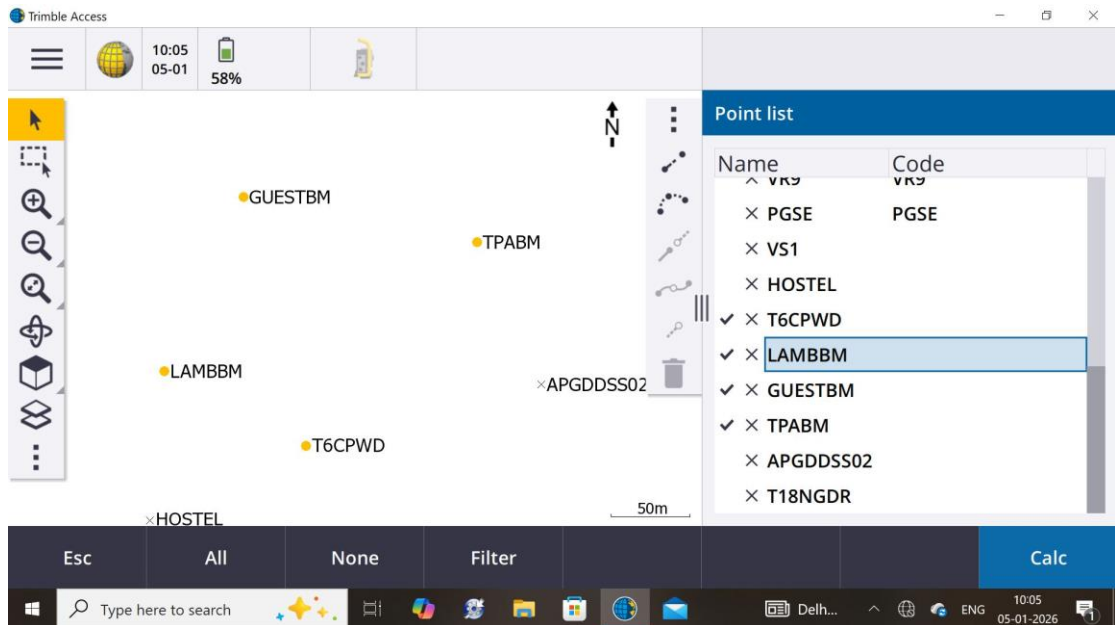
For area calculations using observed points (via NRTK/RTK/RTX) we have to follow below steps:-

- Open Project and Job then Click >> **Menu button** >> **Cogo** >> **Area calculations**

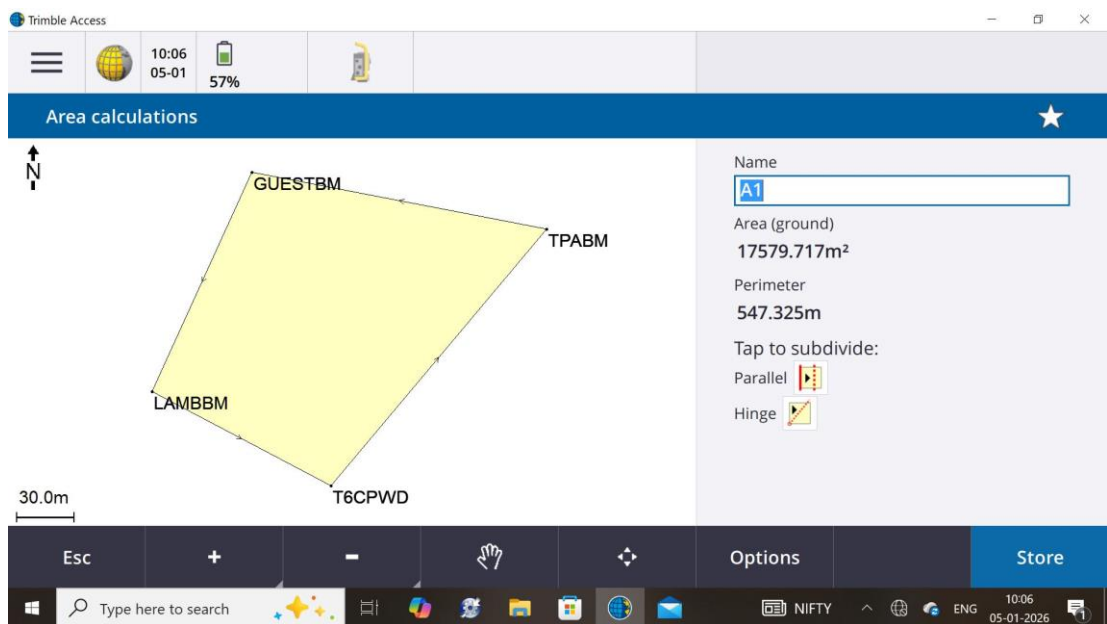


- A list of observed points will appear in the right side, we have to select all corner points of the desired area either in clockwise or in anti-clockwise manner.





- Click >> **Measure** and area would be shown like as below, we can save this by assigning a name.

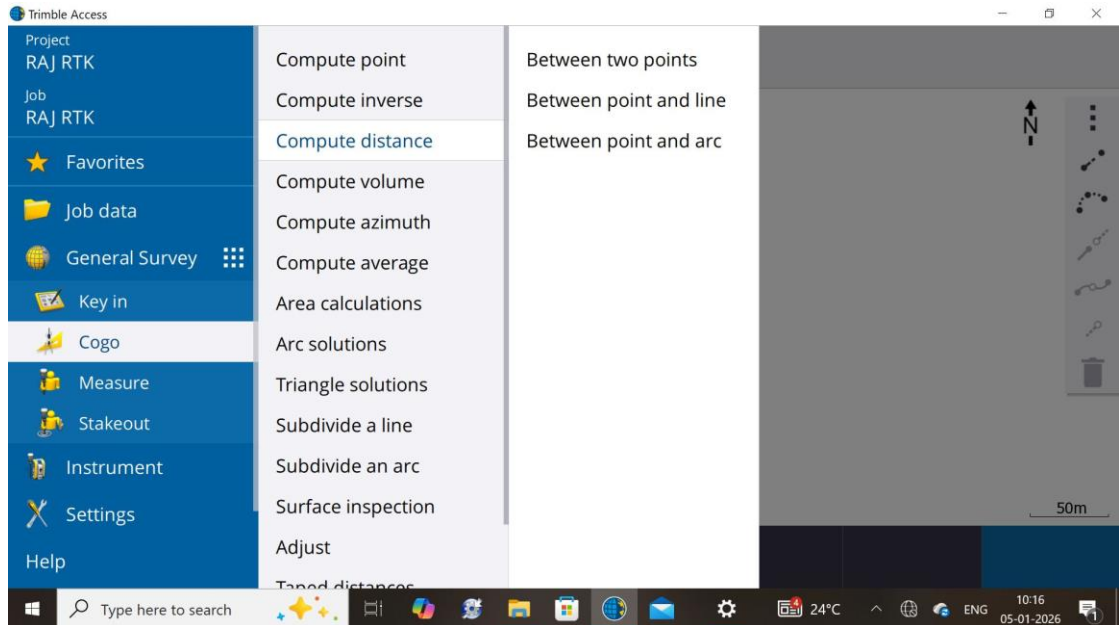


- Click >> **Store**

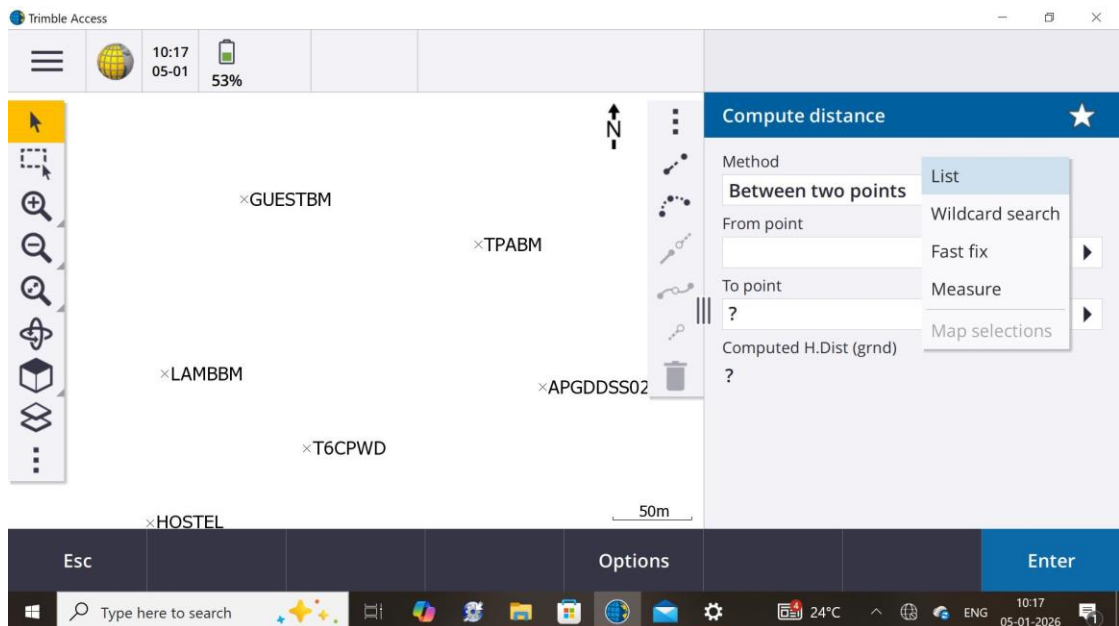
19. Annexure E : Distance Measurement

For Distance measurements using observed points (via NRTK/RTK/RTX) we have to follow below steps:-

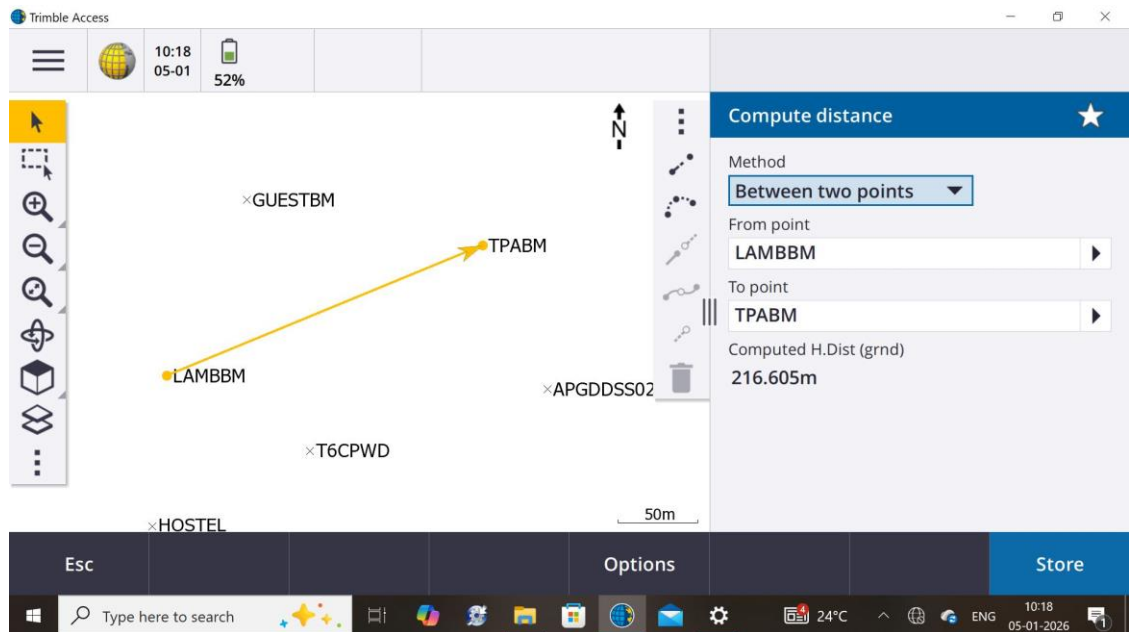
- Open Project and Job then Click >> **Menu button** >> **Cogo** >> **Compute distance** >> **Between two points/Between point and line**



- A box will appear in the right side as follows:



- Select method of computing distance e.g. Between two points and afterwards select two desired points



- Click >> **Store**